



**LLM in Court: Leveraging Open-Source Language Models to Enhance
Empirical Legal Research in Brazil**

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Master of Data Science for Public Policy, Class of 2025

Berlin, April 28, 2025

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Word count: 7,000 words

*"I promise to practice law with dignity and independence, to observe ethics, professional duties and prerogatives and to defend the Constitution, the legal order of the Democratic State, human rights, social justice, the proper application of laws, the rapid administration of justice and **the improvement of legal culture and institutions.**"*

(Oath of the Legal Profession - Art. 20, General Regulation of the Statute of the Lawyer Professorship, Brazilian Bar Association)

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Executive Summary

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Introduction

- context:
 - law is a fundamental aspect in governance and, therefore, to well being
 - however, law is heavily dogmatic and there is little empirical legal research in the country.
 - to do empirical research on law the data mining process is specially complicated, not because the lack of material – Judiciary in BR is full digitalized – but to analyze a large volume of data, with it's specific linguo
 - the first obstacle to a data-driving analysis in law is how to manage text data in large scale and extract structured information from it: one could easily take months to properly analyze few dozens of complex documents.
 - one side-hipothesis: our legal studies and legislative production is suboptimal, due the lack of evidence based decisions
 - relevant to a policy school; in a data science program it's specially relevant to point out solutions to improve this field and offer policy recommendations in the light of the state of the art
- contribution: pipeline, analyse, at least 1 os model trained

There is a significant methodological gap in legal research. Empirical methods are rarely applied, resulting in predominantly qualitative, bibliographical research. When empirical research is done in the legal field, it usually takes a substantial period and a big research team, making it a costly process.

Since the most challenging part of empirical analysis in the legal field is extracting informa-

tion from very complex documents, I hope to develop a good strategy to make it easier for researchers to analyze how the law is being applied by the courts. Also, I want to apply open-source models since they are a better fit for the public interest and are most easily adoptable by academia and public institutions.

Literature Review

- literature: lawma, tucano, marina lacerda, rayssa, abj drogas Dominguez-Olmedo et al. (2024), Corrêa et al. (2024), Lacerda e Silva (2021), Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023), Associação Brasileira de Jurimetria (2019), Costa et al. (2011)

The primary focus is the lack of empirical research in the legal field and the costly, time-consuming nature of conducting such studies. While traditional legal research, like that of the Brazilian Applied Economics Institute, and jurimetrics research using data-driven techniques exist, they are limited. In contrast, recent advancements in the application of LLMs to the legal field in the U.S. (e.g., Dominguez-Olmedo et al., Lawma: The Power of Specialization for Legal Tasks) show promise. My thesis will explore the role of open-source models, techniques for their specialization in legal tasks, and the costs of their deployment.

Research Question

At one hand, we recognize a research gap regarding the lack of empirical legal research mainly in Brazil. On the other hand, we see an opportunity to apply the findings of the fast-paced field of natural language processing, empowered by the modern technichs from deep learning. Giving this setting, the research questions relies on: 1. How can large language models be deployed in the Brazilian legal context to accurately extract information, recognize entities, and summarize complex legal documents, considering factors such as cost, computational

efficiency, and the specific challenges of legal language and document structure?

This question implies: - is possible to use commercial llms in similar tasks - the new releases on open-source models might reached a point when those are also a perspective

Research Design

- method:
 - gathered data from judicial sentences in São Paulo
 - received access from dataset from Lacerda: $\pm$ 400 judicial sentences labelled
 - cleaned data
 - prompt engeneering
 - used commercial OS (OpenAI) as baseline
 - tryied the OS models
 - * llama
 - * gemma
 - * tucano
 - * lawma
 - * lola
 - libraries HF and unsloth
 - ft models
 - compared their performance
 - limitations:
 - * drug traffic, sao paulo
 - * 2017/2018
 - * only models that were possible to access
 - * small sample to ft and test
 - * tucano, lawma:

- context window was too short to legal docs;
- tried with no success to apply rope scaling or yarn
- tried RAG to retrieve relevant parts; tried rag to squish the text
- possibility: try a task-based prompt – still limited window, but might be sufficient to many documents
- lack of computational resources

Data Sources

- data from legal experts – Lacerda e Silva ([2021](#))

Methods

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Limitations

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Results

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Discussion

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Key Findings

- results - os: - cons: difficult to manage, several have short context window; computationally intensive; - pro: privacy; costs; fast paced; adaptability; full control; - commercial: - cons: privacy; costs; - pro: reliability; easiness; credibility - indeed, specialization is a good strategy - have good potential, although more testing is needed

Future Research

- future research:
 - quantization; pruning; distillation;
 - task-based approach

Policy Recommendations

- policy recs:
 - academia: open the data from the empirical studies -> corpus of legal label
 - os community: larger context windows, more light models
 - judiciary branch: improve metadata recollection and open access to judicial decisions

Conclusion

- * conclusions:
 - llms are good tools to leverage empirical legal research in BR
 - commercial models: still up to date a good strategy, despite of its costs and privacy
 - pre-process to anonymize is an option
 - os models: feasible and decent baseline performances in certain models; ft is a good

Alva-Manchego et al. ([2021](#))

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References

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Alva-Manchego, F., Scarton, C., & Specia, L. (2021). The (Un)Suitability of Automatic Evaluation Metrics for Text Simplification. *Computational Linguistics*, 47, 861–889. https://doi.org/10.1162/coli_a_00418

Associação Brasileira de Jurimetria. (2019). *Avaliação do Impacto de Critérios Objetivos na Distinção Entre Posse para Uso e Posse para Tráfico: Um estudo Jurimétrico*. Associação Brasileira de Jurimetria. https://abj.org.br/pdf/20190402_abj_criterios_objetivos.pdf

Corrêa, N. K., Sen, A., Falk, S., & Fatimah, S. (2024). *Tucano: Advancing Neural Text Generation for Portuguese*. arXiv. <https://doi.org/10.48550/arXiv.2411.07854>

Costa, S. H. da, Silva, P. E. A. da, Sabino, M. A. da C., Fernandes, D. C. M., & Lusvarghi, L. A. dos S. (2011). *A eficácia do sistema jurídico de prevenção e combate à improbidade administrativa* [Publicação de relatório de pesquisa]. Secretaria de Assuntos Legislativos do Ministério da Justiça.

Dominguez-Olmedo, R., Nanda, V., Abebe, R., Bechtold, S., Engel, C., Frankenreiter, J., Gummadi, K., Hardt, M., & Livermore, M. (2024). *Lawma: The Power of Specialization for Legal Tasks*. arXiv. <http://arxiv.org/abs/2407.16615>

Instituto de Pesquisa Econômica Aplicada (Ipea), & Brasil. Ministério da Justiça e Segurança Pública. (2023). *Perfil do processado e produção de provas nas ações criminais por tráfico de drogas : Relatório analítico nacional dos Tribunais Estaduais de Justiça Comum*. Instituto de Pesquisa Econômica Aplicada (Ipea). <https://doi.org/10.38116/ri221151>

Lacerda e Silva, M. (2021). *Punindo as diferenças: Gênero, raça e geração no sentenciamento de tráfico de drogas na cidade de São Paulo*. <http://repositorio.unb.br/handle/10482/41413>

Acknowledgements

Marina Lacerda Silva, Lynn Kaack, Huy Dang, Sascha. Prof. Susana Henriques da Costa, José Jesus Filho, Associação Brasileira de Jurimetria e Jusbrasil. MDS classes of 2023 and 2025, the Latin-American Club specially the Brazilians. Special thanks to João Vitor Krieger, Renan Barbosa. Also, a grateful to Fernanda Horn, Vitor Moneo, Orisha, Meirelles.

Appendix

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List of experiments